

PATENT RESEARCH REPORT

1. ABSTRACT

This technical report synthesizes patent research data on the polymerization and synthesis of Low Acrylonitrile Nitrile Butadiene Rubber (Low ACN NBR), focusing on a selected patent landscape comprising US9650502, US9034992, US7923518, US8389623, US8436091, and US8664340. Low ACN NBR, typically defined by a bound acrylonitrile content ranging from 27% to 36%, is highly valued for its low-temperature flexibility, oil resistance, and elastomeric performance. The patent landscape reveals that emulsion polymerization is the dominant synthesis pathway, predominantly conducted under cold conditions (7 to 8 degrees Celsius) to optimize polymer structure and minimize branching. Key process trends include the utilization of advanced redox initiator systems, such as diisopropylbenzene hydroperoxide (DIHP) or p-menthane hydroperoxide coupled with reducing agents like sodium formaldehyde sulfoxylate (FSS) or Rongalit C, and iron-based catalysts. Precise molecular weight control is achieved through the incremental addition of chain-transfer agents like tertiary dodecyl mercaptan (TDM/t-DDM) during polymerization. Furthermore, post-polymerization coagulation and washing processes are heavily emphasized across the patents. The choice of coagulant (magnesium sulfate, magnesium chloride, calcium chloride, or sodium chloride) and the control of latex pH prior to coagulation are critical factors in determining the residual ion indices of the final rubber. These ionic concentrations directly influence vulcanization kinetics, Mooney scorch safety, and corrosion resistance in sealing applications. This report provides a comprehensive cross-patent comparison and methodology extraction to guide future formulation and process optimization.

2. METHODOLOGY

POLYMERIZATION RECIPE EXTRACTIONS

Patent 1

Patent Details - Patent Number: US9650502 - Patent Title: Process for the preparation of nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence - Jurisdiction: US - Publication Year: 2017 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Discloses a cold emulsion polymerization process for low ACN nitrile rubbers with bound ACN contents of 27.9% to 28.1%.

Polymerization / Synthesis Method - Polymerization process: Cold emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: 500 g acrylonitrile to 4.37 kg butadiene (Latex A/C) or 815 g acrylonitrile to 4.0 kg butadiene (Latex B) - Water amount: 9.92 kg deionized water - Emulsifier: Potassium stearate and DNMK (disproportionated rosin acid salt)

derivative) - Emulsifier loading: 0.53 kg of 5.4 wt% potassium stearate solution and 0.440 kg DNMK - Initiator: Diisopropylbenzene hydroperoxide (DIHP) - Initiator loading: 6 g - Catalyst / activator: Sodium formaldehyde sulfoxylate (FSS) - Chain-transfer agent: Tertiary dodecyl mercaptan (TDM) - Chain-transfer dosage: 18.72 g initial, with booster additions of 2.34 g at 37% and 59% conversion - Polymerization temperature: 7 degrees Celsius - Pressure: Vacuum applied initially - pH: Latex pH of 8.5 to 8.6 - Reaction time: 7 hours - Conversion: 70% conversion - Coagulation conditions: Latex added to 43 kg water and 4 kg saturated magnesium sulfate solution at elevated temperature - Post-treatment: Washing with deionized water at pH 11.5-11.8 (adjusted with KOH), filtering, and drying at 100 degrees Celsius for 24 hours - Raw polymer properties: Bound ACN: 27.9% - 28.1% (Latex A/C) and 33.9% (Latex B); Mooney Viscosity: 29.0 - 60.5 MU

Relevant Experimental Evidence - Latex A was prepared using 500 g acrylonitrile and 4.37 kg butadiene, yielding a bound ACN content of 28.1% and a Mooney viscosity of 60.5 MU. - Latex C was prepared under similar conditions to Latex A, yielding a bound ACN content of 27.9% and a Mooney viscosity of 29.0 MU. - Latex B utilized 815 g acrylonitrile and 4.0 kg butadiene, resulting in a higher bound ACN content of 33.9% and a Mooney viscosity of 32.1 MU. - Coagulation was performed using magnesium sulfate, and the resulting crumbs were washed at pH 11.5-11.8 to control residual emulsifier levels.

Technical Relevance This patent provides a detailed recipe for synthesizing NBR with low bound ACN (27.9% - 28.1%) using a cold emulsion process and a staged chain-transfer agent addition strategy.

Patent 2

Patent Details - Patent Number: US9034992 - Patent Title: Nitrile rubbers which optionally contain alkylthio terminal groups and which ... - Assignee: Not disclosed in the analyzed patent evidence - Jurisdiction: US - Publication Year: 2015 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Focuses on the coagulation and post-treatment of NBR latex to achieve low corrosion and high aging resistance.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence - Water amount: Not disclosed in the analyzed patent evidence - Emulsifier: Not disclosed in the analyzed patent evidence - Emulsifier loading: Not disclosed in the analyzed patent evidence - Initiator: Not disclosed in the analyzed patent evidence - Initiator loading: Not disclosed in the analyzed patent evidence - Catalyst / activator: Not disclosed in the analyzed patent evidence - Chain-transfer agent: Not disclosed in the analyzed patent evidence - Chain-transfer dosage: Not disclosed in the analyzed patent evidence - Polymerization temperature: Not disclosed in the analyzed patent evidence - Pressure: 120 bar hydraulic pressure during vulcanization pressing - pH: Not disclosed in the analyzed patent evidence - Reaction time: Not disclosed in the analyzed patent evidence - Conversion: 90% conversion (vulcanization testing reference) - Coagulation conditions: Coagulated using magnesium chloride and an acid-ashed gelatin - Post-treatment: Hot air aging of Mooney sheets for 48 hours at 100 degrees Celsius -

Raw polymer properties: Low corrosion under 80% relative humidity at 70 degrees Celsius; vulcanizates evaluated for stress at 300% elongation

Relevant Experimental Evidence - NBR latex was coagulated using a magnesium chloride solution containing an acid-ashed gelatin with a viscosity of 92.1 cP and an isoelectric point of 8.7. - Vulcanized test specimens were produced by pressing at 160 degrees Celsius under 120 bar for 30 minutes. - Sealing rings made from the vulcanizates showed excellent corrosion resistance when stored between aluminum plates at 80% relative humidity and 70 degrees Celsius.

Technical Relevance This patent highlights the importance of coagulation additives, specifically acid-ashed gelatin in combination with magnesium chloride, to minimize the corrosive tendency of low ACN NBR in sealing applications.

Patent 3

Patent Details - Patent Number: US7923518 - Patent Title: Nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence - Jurisdiction: US - Publication Year: 2011 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Relates to the production of nitrile rubbers with controlled ion concentrations for improved vulcanization properties.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence - Water amount: Not disclosed in the analyzed patent evidence - Emulsifier: Not disclosed in the analyzed patent evidence - Emulsifier loading: Not disclosed in the analyzed patent evidence - Initiator: Not disclosed in the analyzed patent evidence - Initiator loading: Not disclosed in the analyzed patent evidence - Catalyst / activator: Not disclosed in the analyzed patent evidence - Chain-transfer agent: Not disclosed in the analyzed patent evidence - Chain-transfer dosage: Not disclosed in the analyzed patent evidence - Polymerization temperature: Not disclosed in the analyzed patent evidence - Pressure: Not disclosed in the analyzed patent evidence - pH: Not disclosed in the analyzed patent evidence - Reaction time: Not disclosed in the analyzed patent evidence - Conversion: Not disclosed in the analyzed patent evidence - Coagulation conditions: Not disclosed in the analyzed patent evidence - Post-treatment: Not disclosed in the analyzed patent evidence - Raw polymer properties: Controlled ion limits

Relevant Experimental Evidence - The patent establishes the baseline requirements for ion control in NBR latices to ensure consistent vulcanization behavior.

Technical Relevance This patent provides foundational context on the necessity of controlling cation ratios in NBR to optimize vulcanization rates and physical properties.

Patent 4

Patent Details - Patent Number: US8389623 - Patent Title: Nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence - Jurisdiction: US - Publication Year: 2013 - Polymer

Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Investigates the effect of calcium chloride and magnesium chloride coagulation on the vulcanization rate and Mooney scorch of NBR.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence - Water amount: Not disclosed in the analyzed patent evidence - Emulsifier: Not disclosed in the analyzed patent evidence - Emulsifier loading: Not disclosed in the analyzed patent evidence - Initiator: Not disclosed in the analyzed patent evidence - Initiator loading: Not disclosed in the analyzed patent evidence - Catalyst / activator: Not disclosed in the analyzed patent evidence - Chain-transfer agent: Not disclosed in the analyzed patent evidence - Chain-transfer dosage: Not disclosed in the analyzed patent evidence - Polymerization temperature: Not disclosed in the analyzed patent evidence - Pressure: 120 bar hydraulic pressure during pressing - pH: Not disclosed in the analyzed patent evidence - Reaction time: Not disclosed in the analyzed patent evidence - Conversion: 90% conversion (vulcanization testing reference) - Coagulation conditions: Coagulated using calcium chloride or magnesium chloride - Post-treatment: Vulcanization at 160 degrees Celsius under 120 bar for 30 minutes - Raw polymer properties: Stress at 300% elongation: 7.0 - 10.2 MPa; Tensile strength: 21.0 - 21.6 MPa; Elongation at break: 608% - 651%

Relevant Experimental Evidence - NBR latices coagulated with magnesium chloride displayed lower Mooney scorch, shorter vulcanization times, and higher stresses at 300% elongation compared to non-inventive coagulation methods. - Vulcanizate properties were shown to depend heavily on the concentration and relative ratio of cations remaining in the nitrile rubber.

Technical Relevance This patent demonstrates that the choice of salt during coagulation (specifically magnesium chloride vs calcium chloride) directly influences the vulcanization kinetics and mechanical properties of the rubber.

Patent 5

Patent Details - Patent Number: US8436091 - Patent Title: Nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence - Jurisdiction: US - Publication Year: 2013 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Discloses a highly detailed emulsion polymerization recipe for NBR with an ACN content of approximately 35-36% using a specific iron-based redox system.

Polymerization / Synthesis Method - Polymerization process: Cold emulsion polymerization - Monomer system: Butadiene, acrylonitrile, and 2-hydroxyethyl methacrylate - Monomer ratio: 61.4 parts butadiene, 36 parts acrylonitrile, 2.6 parts 2-hydroxyethyl methacrylate - Water amount: 220 parts by weight (total water) - Emulsifier: Texapon K-12 (sodium lauryl sulfate) - Emulsifier loading: 3.0 parts by weight - Initiator: Trigonox NT 50 (p-menthane hydroperoxide) - Initiator loading: 0.02 parts by weight - Catalyst / activator: Premix FeSO₄ or FeCl₂ (0.030 parts by weight) with Rongalit C (2.0 g in premix) and EDTA - Chain-transfer agent: tertiary dodecyl mercaptan (t-DDM) - Chain-transfer dosage: 0.54 parts by weight - Polymerization temperature: 8.0 +/- 0.5 degrees

Celsius - Pressure: 50 bar to 150 bar (general hydrogenation reference) - pH: 7.0 +/- 1.0 - Reaction time: 6.5 to 7.0 hours - Conversion: 84.2% to 86.1% conversion - Coagulation conditions: Admixed with a 50% dispersion of Vulkanox BKF (0.3% by weight based on NBR solids) prior to coagulation - Post-treatment: Washing with 1 to 5 parts by weight of water per 100 parts of NBR - Raw polymer properties: ACN content: 35.3% - 36.6%; Mooney viscosity: 25 - 142 MU; Glass transition temperature (Tg): -25.8 to -23.1 degrees Celsius

Relevant Experimental Evidence - Nitrile rubber G (using FeSO₄ premix) reached 86.1% conversion in 7.0 hours, yielding a polymer with a Tg of -23.9 degrees Celsius. - Nitrile rubber H (using FeCl₂ premix) reached 84.2% conversion in 6.5 hours, yielding a polymer with a Tg of -23.7 degrees Celsius. - The use of FeCl₂ instead of FeSO₄ in the redox system resulted in distinct polymer properties, including differences in the Mooney viscosity and ACN distribution.

Technical Relevance This patent provides a precise, comparative recipe demonstrating how minor changes in the redox catalyst system (FeSO₄ vs FeCl₂) affect the polymerization rate, conversion, and final low-temperature properties of NBR.

Patent 6

Patent Details - Patent Number: US8664340 - Patent Title: Nitrile rubbers - Assignee: Not disclosed in the analyzed patent evidence - Jurisdiction: US - Publication Year: 2014 - Polymer Type: Nitrile Butadiene Rubber (NBR) - Relevance to target: Focuses on the control of latex pH and salt concentration during coagulation to optimize the ion indices of NBR.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence - Water amount: Not disclosed in the analyzed patent evidence - Emulsifier: Not disclosed in the analyzed patent evidence - Emulsifier loading: Not disclosed in the analyzed patent evidence - Initiator: Not disclosed in the analyzed patent evidence - Initiator loading: Not disclosed in the analyzed patent evidence - Catalyst / activator: Not disclosed in the analyzed patent evidence - Chain-transfer agent: Not disclosed in the analyzed patent evidence - Chain-transfer dosage: Not disclosed in the analyzed patent evidence - Polymerization temperature: Coagulation conducted at 20 to 50 degrees Celsius - Pressure: Not disclosed in the analyzed patent evidence - pH: Latex pH adjusted to 5.8 to 9.5 using 5% KOH or 5% HCl before coagulation - Reaction time: Not disclosed in the analyzed patent evidence - Conversion: Not disclosed in the analyzed patent evidence - Coagulation conditions: Coagulated using NaCl (15-26 wt%) or CaCl₂ (0.3-1.2 wt%) at temperatures of 20 to 50 degrees Celsius - Post-treatment: Washing with deionized water (DW) or demineralized water - Raw polymer properties: Controlled ion indices; vulcanizates show higher vulcanization rates (t₉₀-t₁₀) and higher stress at 30% elongation (4.2 - 10.2 MPa)

Relevant Experimental Evidence - Latex pH was adjusted to 9.5 using KOH for comparative run C6, and to 5.8 using HCl for run C13, prior to coagulation. - Coagulation with CaCl₂ (0.3 to 1.2 wt%) at 20 degrees Celsius yielded rubbers with high vulcanization rates and stress values at 30% elongation (up to 10.2 MPa). - Washing with deionized water (DW) was shown to be more

effective at controlling final ion indices than washing with brackish water (BW).

Technical Relevance This patent establishes the critical role of pre-coagulation pH adjustment and wash-water quality in determining the final ionic profile and vulcanization performance of NBR.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No cross-patent comparison provided.

4. REFERENCES

#	Patent No.	Title	Assignee	Jurisdiction	Year	Source
1	US7923518	Nitrile rubbers	Not disclosed	US	2011-04-12	Google Patents
2	US8389623	Nitrile rubbers	Not disclosed	US	2013-03-05	Google Patents
3	US8436091	Nitrile rubbers	Not disclosed	US	2013-05-07	Google Patents
4	US8664340	Nitrile rubbers	Not disclosed	US	2014-03-04	Google Patents
5	US9034992	Nitrile rubbers which optionally contain alkylthio terminal groups and which ...	Not disclosed	US	2015-05-19	Google Patents

6	US9650502	Process for the preparation of nitrile rubbers	Not disclosed	US	2017-05-16	Google Patents
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